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CSE428

section 03

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Submission date : 04.04.2026

Problem Set 1

Lecture 1

(1)

Image 1 → Absorption

Image 2 → Emission

Image 3 → Reflection

(2) Aspect ratio = 5 : 3

Total pixels = 960000 pixels

Now, $5x * 3x = 960000$

$\Rightarrow 15x^2 = 960000$

$\Rightarrow x = 252.98 \approx 253$

Dimension of the image = $(5 \times 253,\ 3 \times 253)$

$= (1265, 759)$

Size of the image = 960000×8

$= 7680000$

(3)

1D signal	2D signal	3D signal
1 variable	2 variables	3 variables
1D array	2D Matrix	3D array
Example: Audio	Example: Image	Example: video
Visualization: Wave	Visualization: Image	Voisualization: Volume
Used in: time series data	Used in: Image processing	Used in: Video processing

(4) Pixel value for $f(1, 3) = 0$

(5)

Each pixel's intensity is represented by 4 bits of data. So $2^4 = 16$ possible intensity levels.

12	15	5
15	0	9
9	12	15

Lecture 2

1

$$f(x, y) = i(x, y) \cdot r(x, y)$$

$$\Rightarrow 490 = 700 \cdot r(x, y)$$

$$\Rightarrow r(x, y) = \frac{490}{700} = 0.7$$

Range of $r(x, y)$: $0 <= r(x, y) <= 1$

Range of $f(x, y)$: $0 <= f(x, y) <= \infty$

2 X-ray images are inverted to improve visibility based on brightness discrimination and the weber Ratio ($\Delta I / I$).

In the original X-ray, important details often lie in dark regions (low intensity I), where the weber ratio is large (small differences). Human eye has poor ability to detect small differences. After inversion, those dark regions become bright (high I), making the weber ratio smaller which has better brightness discrimination. where the human eye detects contrast better, improving overall image clarity.

(3) Stages of pipeline:

i) Image Formation: Light from a source reflects off the object and passes through the camera lens, forming a continuous optical image on the sensor plane.

ii) Sensing: The camera sensor captures this light and converts it into an analog electrical signal, where the output is proportional to light intensity.

iii) Digitization: Converts the analog signal into digital form:

Sampling: Divides the image into a grid of pixels.
Quantization: Assigns discrete intensity values.

iv) Representation: The final digital image is stored as $f(x, y)$, a 2D matrix of pixel values, where each value represents intensity.

(4) Dynamic Range: It is the ratio of the maximum intensity to minimum intensity in an image.

High Dynamic Range is desirable:

i) Captures details in both dark and bright regions.
ii) Prevents loss of information due to clipping or saturation.
iii) Produces images that are more realistic and visually informative.

Image (b) has a higher dynamic range. In image a, the bright snow and sky are overexposed (clipped at Lmax), causing a complete loss of detail in those bright areas. In contrast, image (b) retains texture in the bright snow while still keeping the darker regions, like the trees, cleanly visible.

(5) Image sensing in a camera is done using a grid like array of thousands of individual sensors called pixels. When light falls on the surface of a sensor through a filter, it generates an electrical response. This response is proportional to the integral of light intensity on the sensor over a period of time, meaning that more light or longer exposure produces a stronger voltage. The signal generated is initially a continuous analog electrical signal, which is later digitized to form the digital image. So, each pixel receives light and produces a voltage signal that represents the brightness of that part of the image.

⑥ Image Digitization is necessary because computers cannot process continuous analog signals, so the continuous voltage from the sensor must be converted into a numerical array.

This process has 2 steps:

Sampling: It is the first step, where the continuous scene is divided into a grid of discrete points or pixels. This step determines the spatial detail and sharpness of the image. Example: a 1024 x 1024 pixel image samples the scene at 1048576 points.

Quantization: This is the 2nd step, where the continuous intensity or voltage at each sample is mapped to a finite set of digital values, such as 0 - 255 for an 8 bit image. This step affects the smoothness and gradation of brightness in the image.

⑦

Image (a): The image shows false contouring with harsh bands instead of smooth gradients due to severe quantization with too few intensity levels. This causes loss of smooth brightness transitions. Using more gray levels solves the problem.

Image (b): The image is pixelated and blocky because of low spatial sampling or too few pixels capturing the scene. This results in loss of fine spatial detail. Capturing at a higher resolution with better interpolation avoids the issue.

(8)

Downsample (2×2):

$$B[i,j] = A[i \cdot k, j \cdot k]$$

$k = 2$, so $B[i,j] = A[i \cdot 2, j \cdot 2]$

$B[0,0] = A[0 \cdot 2, 0 \cdot 2] = A[0,0] = 6$

$B[0,1] = A[0 \cdot 2, 1 \cdot 2] = A[0,2] = 1$

$B[1,0] = A[1 \cdot 2, 0 \cdot 2] = A[2,0] = 1$

$B[1,1] = A[1 \cdot 2, 1 \cdot 2] = A[2,2] = 9$

Downsample Matrix

$$\begin{bmatrix} 6 & 1 \\ 1 & 9 \end{bmatrix}$$

Upsample (8X8)

$$\begin{bmatrix} 6 & 6 & 2 & 2 & 1 & 1 & 9 & 9 \\ 6 & 6 & 2 & 2 & 1 & 1 & 9 & 9 \\ 2 & 2 & 0 & 0 & 2 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 2 & 2 & 4 & 4 \\ 1 & 1 & 9 & 9 & 9 & 9 & 5 & 5 \\ 1 & 1 & 9 & 9 & 9 & 9 & 5 & 5 \\ 2 & 2 & 0 & 0 & 1 & 1 & 2 & 2 \\ 2 & 2 & 0 & 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

(9)

Sensor → Retina

Lens → Optical Lens

Shutter → Iris & Pupil

Focusing region → Fovea

Autofocus system / focus region → ciliary body.

Lecture 3

(1)

image 1

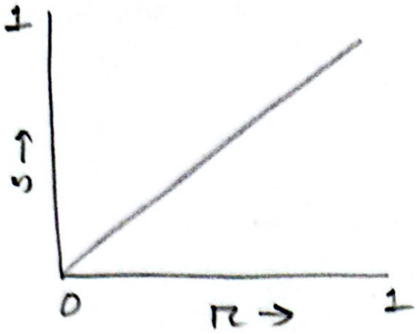


image 2

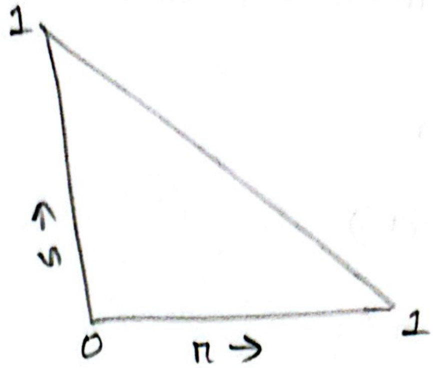


image 4

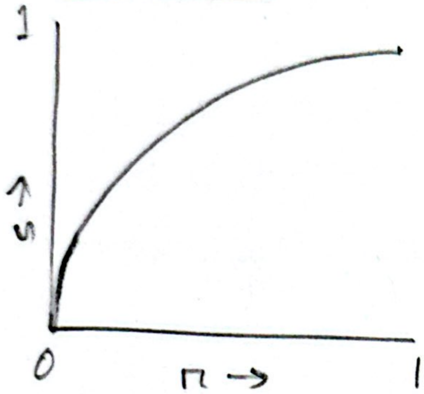
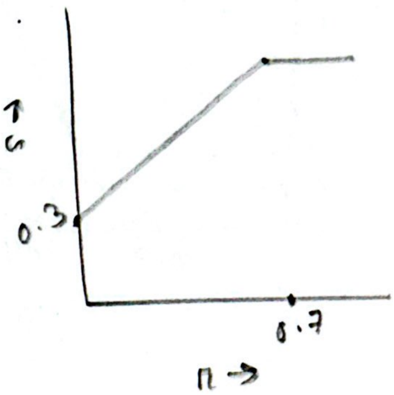


image 3



(2)

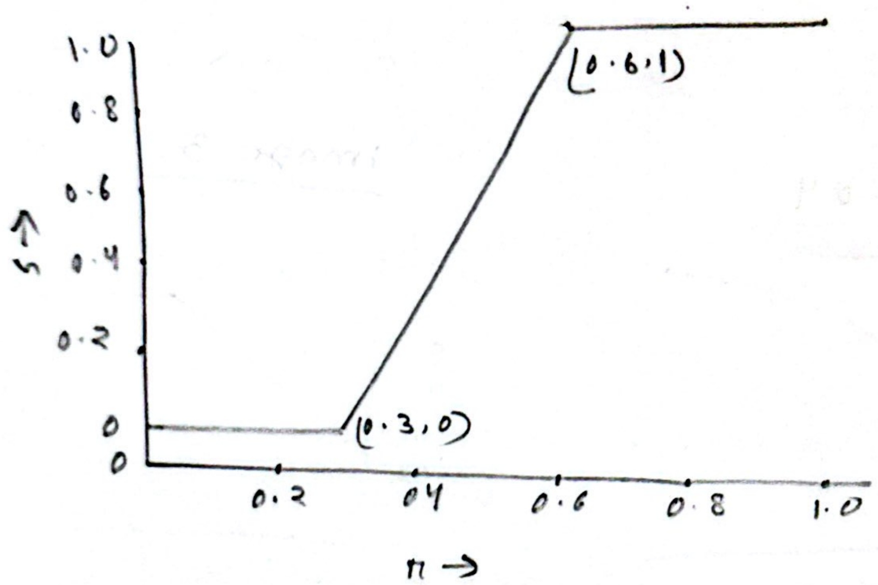
i) Brightness : Brightness increases because adding 6 shifts all pixel intensities up and multiplying by 2 amplifies them, making the image overall brighter.

ii) contrast : Contrast increases as multiplying the pixel values by 2 stretches the intensity range, making dark and light areas more distinct.

iii) data loss: Data loss occurs due to clipping. Pixels exceeding 255 are saturated, causing loss of detail in bright region.

(3)

a)



b) $s = c1(I - c2)$

$0 = c1(0.3 - c2) \Rightarrow c2 = 0.3$

$1 = c1(0.6 - 0.3) \Rightarrow c1 = 3.33$

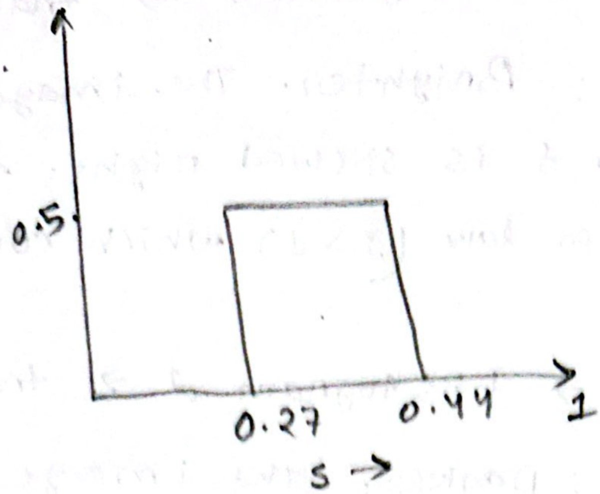
c) Range $(0.3, 0.6)$

$s = \log(1 + 3r)$

Transformed range $(\log(1.9), \log(2.8))$

$= (0.27, 0.44)$

~~x (0.64, 1)~~

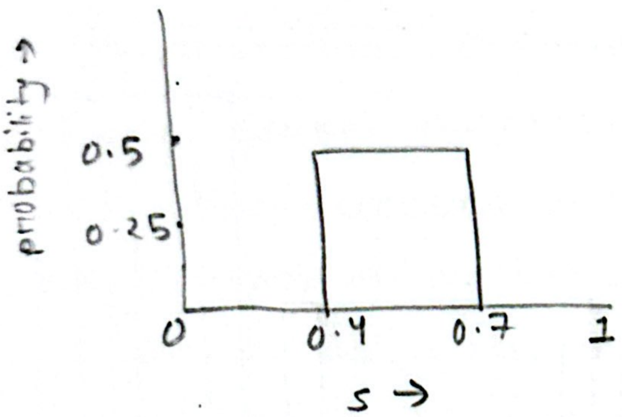


d) $s = 1 - r$

range (0.3, 0.6)

$s = 1 - 0.6 = 0.4$
$s = 1 - 0.3 = 0.7$

Transformed range (0.4, 0.7)



(4) image (a) → histogram (e) → transformation (h)

explanation: Low/Mid contrast. Histogram e is narrow (low contrast). Transformation (h) is a sigmoid curve, which stretches mid-tones to increase contrast.

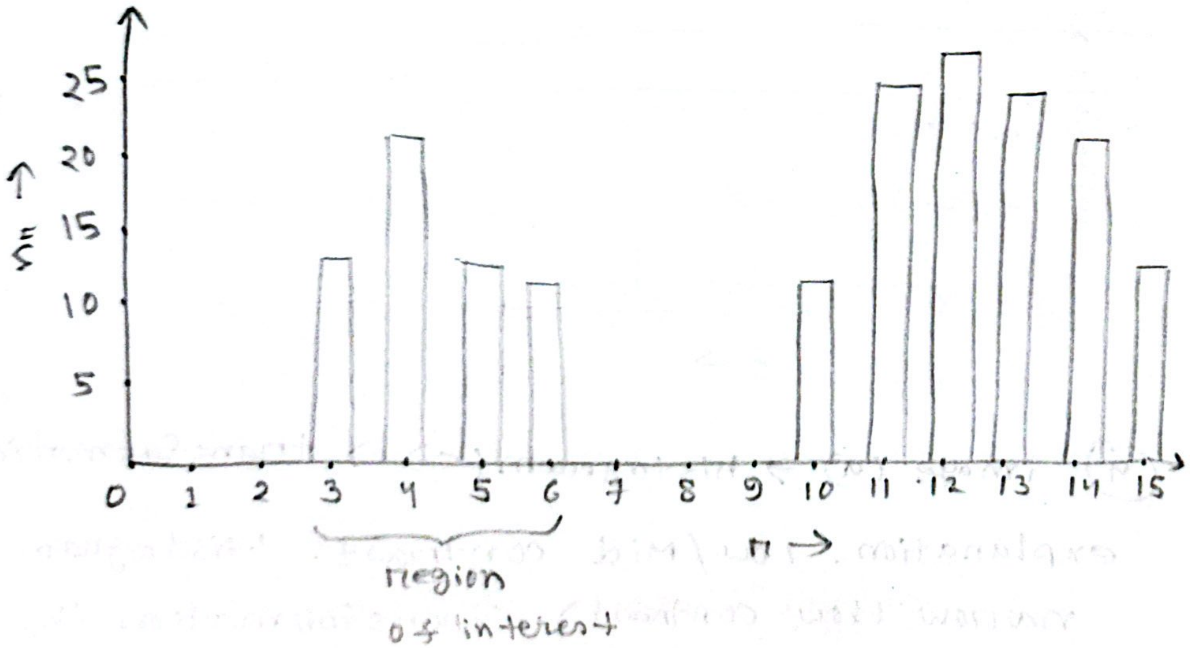
image (b) → histogram f → transformation g.

Explanation : Brighter. The image is washed out. Histogram f is skewed right. Transformation g is a power law ($\gamma > 1$) which curves down.

image (c) → histogram d → transformation i.

Explanation : Darker. The image is in shadow. Histogram d is skewed left (dark). Transformation i is a power law ($\gamma < 1$) which curves up, brightening the dark areas.

(5) a) 3 - 6 → region of interest.

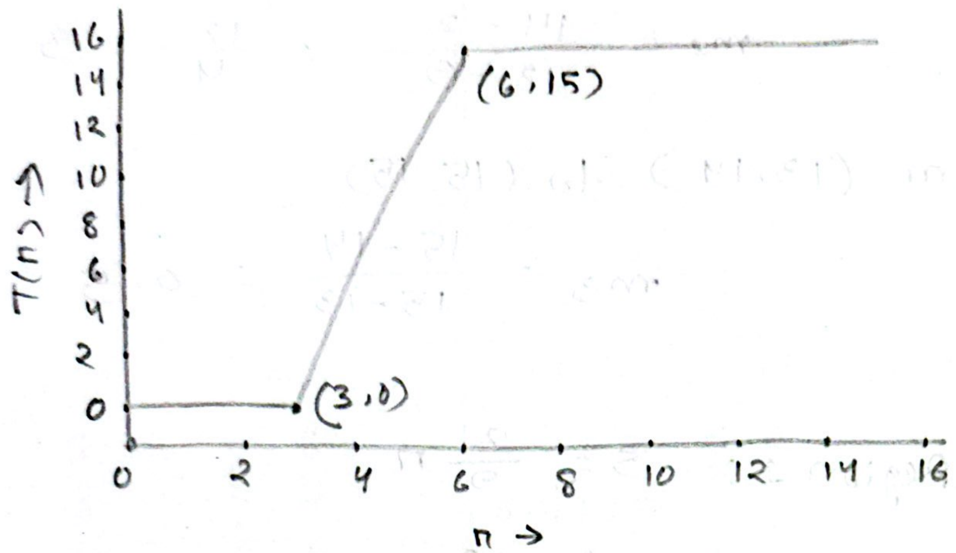


b) $(3,6) \rightarrow (0,15)$

$T(r) = c1(r - c2)$

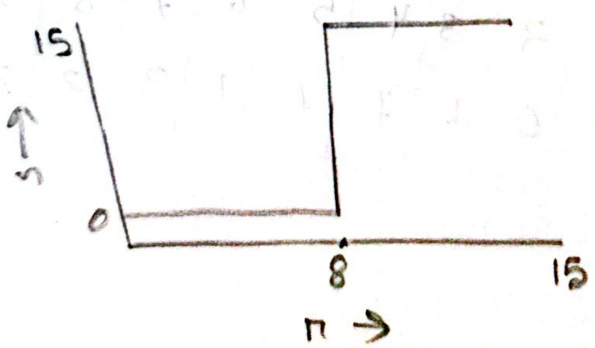
$0 = c1(3 - c2) \Rightarrow c2 = 3$

$15 = c1(6 - 3) \Rightarrow c1 = 5$



c) The transformed image appears darker with thicker, darker numbers and heavier background noise. This occurs because $\gamma > 1$ compresses low-intensity values, pushing them closer to black. As a result, the already dark '428' pixels become even darker, enhancing their visibility.

d)



e) a.

From (0, 0) to (9, 2)

$$m_1 = \frac{2-0}{9-0} = \frac{2}{9} \approx 0.22$$

From (9, 2) to (13, 14).

$$m_2 = \frac{14-2}{13-9} = \frac{12}{4} = 3$$

From (13, 14) to (15, 15)

$$m_3 = \frac{15-14}{15-13} = 0.5$$

b. Region 1: $s = \frac{2}{9} r$

Regim 2: (9, 2)

$$s - 2 = 3(r - 9)$$

$$\Rightarrow s = 3r - 25$$

Regim 3: (13, 14)

$$s - 14 = 0.5(r - 13)$$

$$\Rightarrow s = 0.5r + 7.5$$

c.

input	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
output	0	0	0	1	1	1	1	2	2	2	5	8	11	14	15	15

Lecture 4

(1) a)

r_k	$h(r_k)$	$f_r(r_k)$
0	6	$6/16 = 0.375$
1	7	$7/16 = 0.4375$
2	2	$2/16 = 0.125$
3	1	$1/16 = 0.0625$

b)

r_k	$h(r_k)$	$f_r(r_k)$	$H(r_k)$	$F_r(r_k)$
0	6	0.375	6	0.375
1	7	0.4375	13	0.8125
2	2	0.125	15	0.9375
3	1	0.0625	16	1.00

c) $s_k = round\ (L-1) * F_r(r_k)$

$L = 4$

$r = 0$, $s = round\ (3 * 0.375) = 1$

$r = 1$, $s = round\ (3 * 0.8125) = 2$

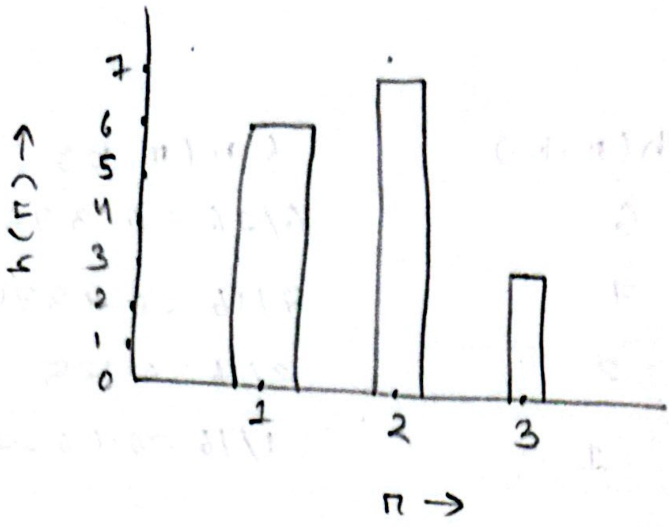
$r = 2$, $s = round\ (3 * 0.9375) = 3$

$r = 3$, $s = round\ (3 * 1.00) = 3$

d)

2	3	1	1
2	1	2	3
1	2	1	2
3	1	2	2

e)



(2) a) $14 \times 14 = 196$ pixel

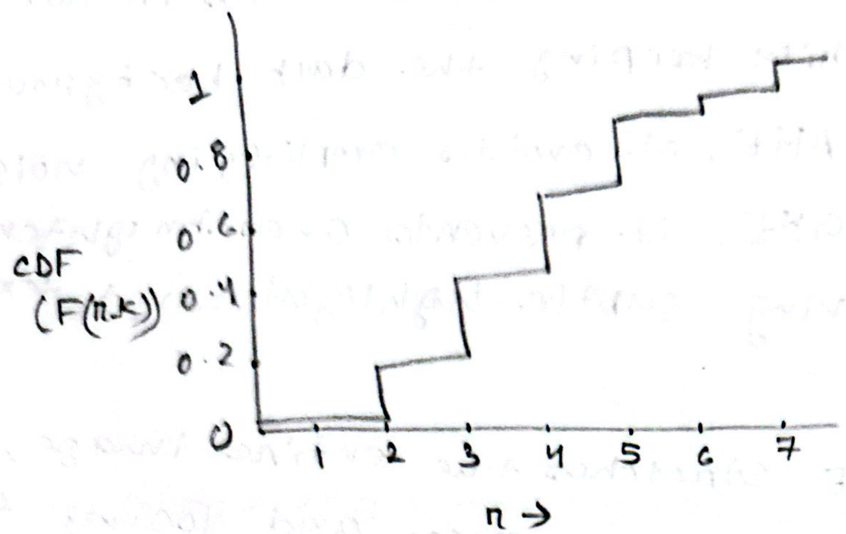
$6 + 0 + 41 + 66 + 42 + 23 + x + 7 = 196$

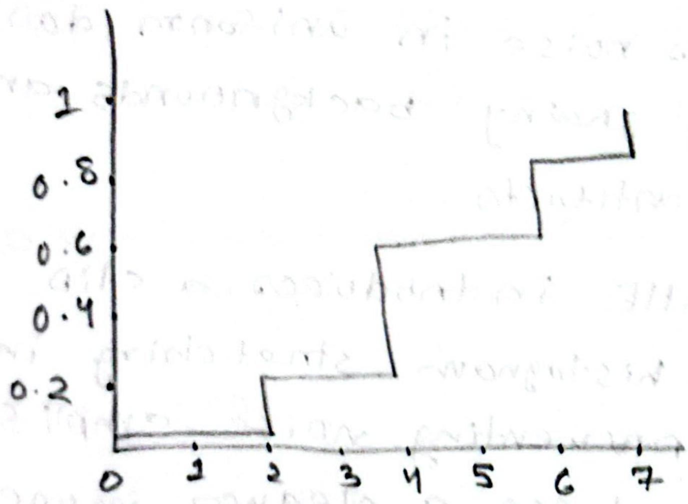
$\Rightarrow x = 11$

b)

n	h(n-k)	f(n-k)	H(n-k)	F(n-k)	HE
0	6	0.03	6	0.03	0
1	0	0	6	0.03	0
2	41	0.2	47	0.23	2
3	66	0.34	113	0.57	4
4	42	0.21	155	0.78	6
5	23	0.12	178	0.90	6
6	11	0.06	189	0.96	7
7	7	0.04	196	1.00	7

c)





In the original image, pixels located at level 2-4, represents low entropy.
After HE, pixels are distributed and spread evenly across 0-8. It increases entropy.

(3)

a) CLAHE enhances local details in the dog's fur while keeping the dark background clean. Unlike AHE, it avoids amplifying noise and unlike GHE, it prevents over brightening, preserving subtle highlights and textures.

b) GHE stretches the entire image, over-brightening dark areas and losing fine details. AHE improves local details but over amplifies noise in uniform dark regions, creating grainy backgrounds and visible tile artifacts.

c) CLAHE introduces a clip limit to restrict histogram stretching in uniform areas, preventing noise amplification. this produces a cleaner more natural image without the grain or halo effects seen in AHE.